

Basic Principles of Medicine I
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 26

Development and Clinical Embryology of the Respiratory System

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Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1428	List the germ layers contributing to the formation of the respiratory tract.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1429	Describe the development of the larynx, trachea, bronchial tree and lungs.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1430	Describe the stages of lung development.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1431	Name the site of production and the role of surfactant.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1432	Summarize the changes that occur in the respiratory system at birth.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1433	Describe the development of the diaphragm.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1434	Describe the division of the embryonic coelom into pleural, pericardial and peritoneal cavities.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1435	Explain the embryological mechanism for all types of congenital diaphragmatic herniae, and eventration.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1436	Explain the embryological mechanism for all types of tracheoesophageal fistulas.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1437	Describe how tracheoesophageal fistula can lead to polyhydramnios.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1438	Explain the embryological basis for laryngeal atresia, tracheal stenosis and atresia, and esophageal atresia and stenosis.

Recommended reading

Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

[Larsen's Human Embryology](#), Ch4, 78-103; Ch11, 245-260 [Link](#)

T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD

[The Developing Human](#), 8, 133-144 [Link](#); Ch10, 183-196 [Link](#)

Pay special attention to the **Green/ blue** boxes for the clinical correlations for this topic.

Images taken from:

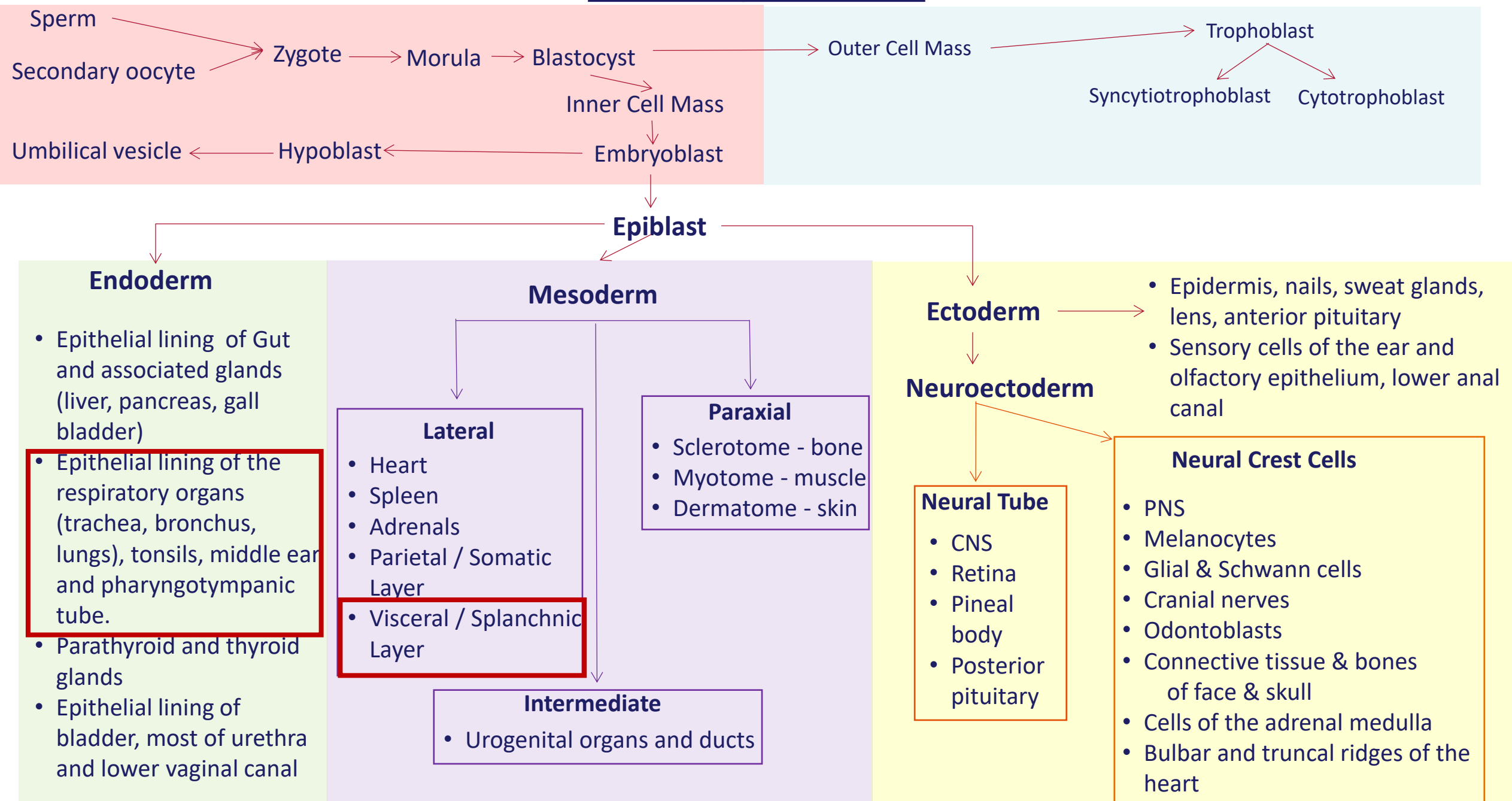
The Developing Human Clinically Oriented Embryology, 11th Edition. Keith L. Moore, T.V.N. Persaud, Mark G. Torchia. *Elsevier*

Langman's Medical Embryology, 14th Edition. T.W. Sadler. *Wolters Kluwer*

Peter H. Abrahams, Jonathan D. Spratt – McMinn's & Abrahams' Clinical Atlas of Human Anatomy, 7th Edition (2013, Mosby).

Atlas of Human Anatomy, 6th Edition – Frank H. Netter, M.D.

The Big picture

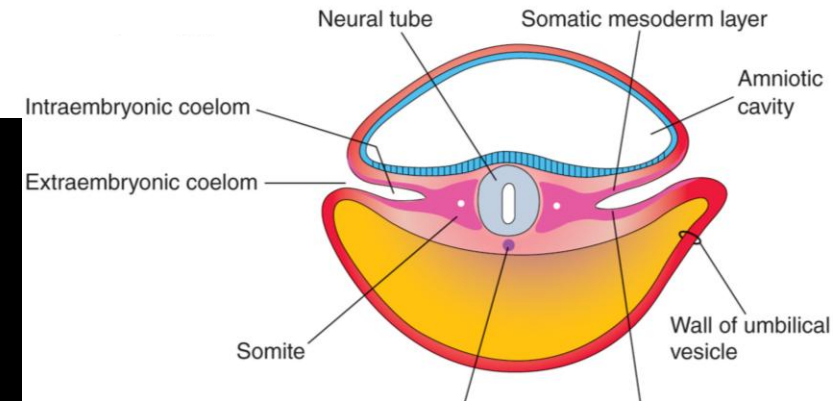
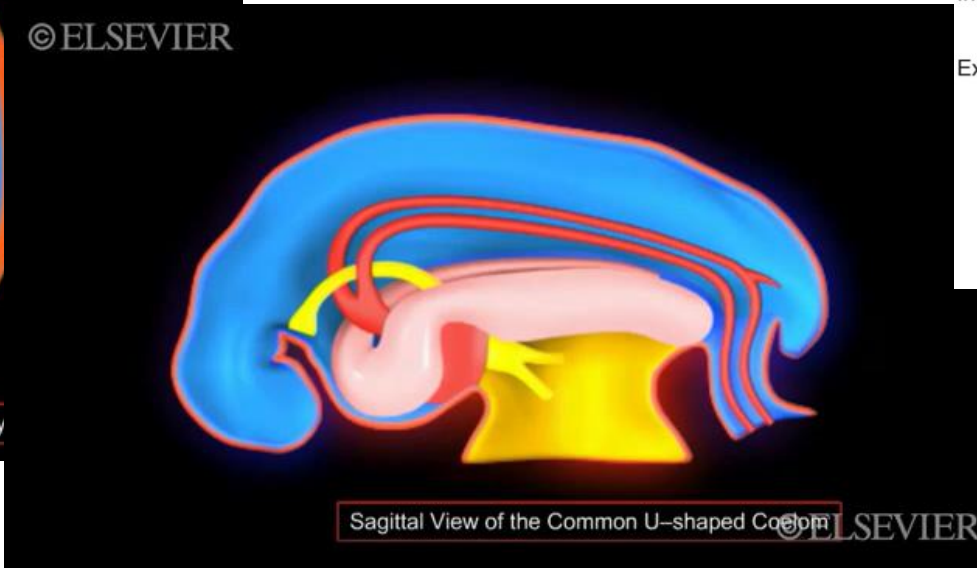
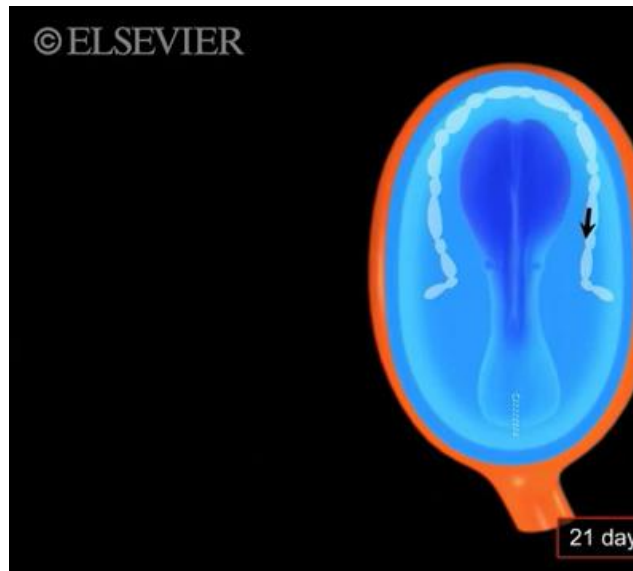


The journey – development of the body cavities and respiratory system

- Development of the intraembryonic coelom – Day 21
 - Cranial end of embryo begins to fold – Day 21
 - Development of the septum transversum – Day 22
 - Development of the respiratory diverticulum – Day 22
 - Caudal end of embryo begins to fold – Day 23
 - Respiratory diverticulum 1st branching gives rise to bronchial buds – Day 26
 - Separation of the respiratory system from the foregut – Day 26 - 28
 - Body folding complete – Day 28
 - Development of the pleuropericardial folds and separation of the pleural and pericardial spaces – Day 28 - 49
 - Secondary bronchial buds form – Day 28 - 30
 - Tertiary bronchial buds and bronchopulmonary segments – Week 7
 - Terminal bronchioles – Week 16
 - Respiratory bronchioles – Week 16 – 26
 - Development of terminal sacs – Week 26 – 39
 - Development of Alveoli – Week 32 till early adulthood
- Week 3 and 4
- Week 5 - 7

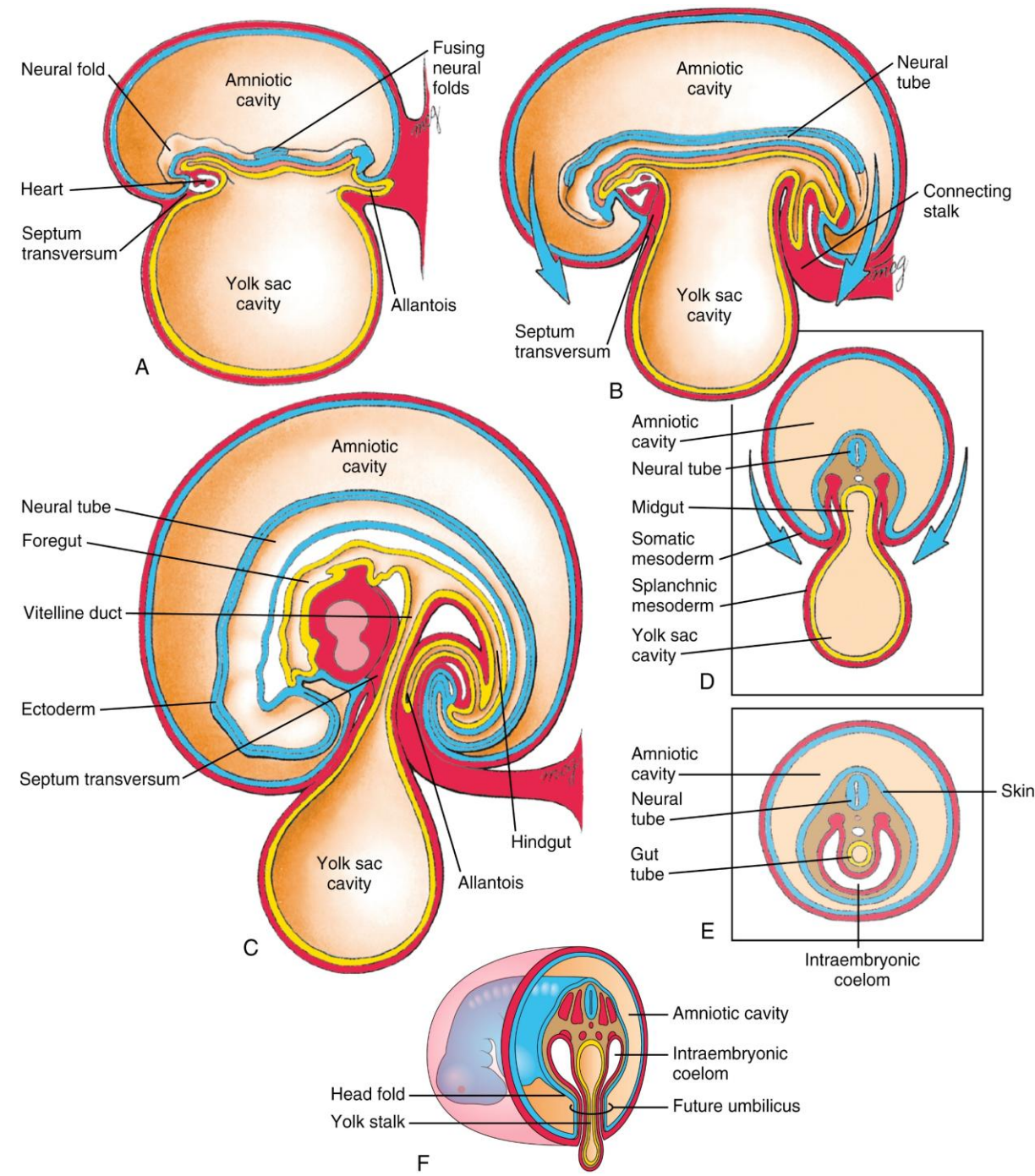
Week 4 – development of the intraembryonic coelom

- **U-shaped space** that develops in the lateral plate mesoderm to form the embryonic body cavities
- Splitting the mesoderm into dorsal **somatic mesoderm** and ventral **splanchnic mesoderm**
 - The space have a parietal wall (close to the somatic mesoderm) and a visceral wall (close to the splanchnic mesoderm)
 - The U bend will become the **pericardial cavity**
 - The limbs will become the **pleural and peritoneal cavities**
 - Distal parts of the limbs are continuous with the extraembryonic coelom



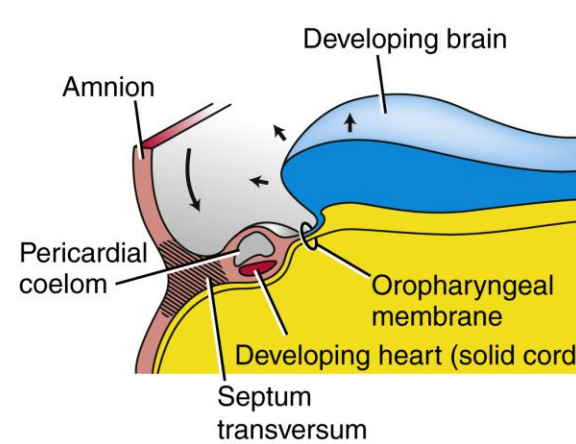
Week 4 - folding

- Caused by differential growth of various tissues
 - **Embryonic disc and amnion grows extensively**
 - Umbilical vesicle hardly any growth
- The developing neural tube, notochord and somites cause the dorsal axis of embryo to stiffen
 - **As these grow they cause the embryo to balloon**
- Cranial (head), caudal (tail) and lateral folds
- Pulled towards each other forming a **ring positioned at the future umbilicus**
 - Forming a tube on top of a tube inside another tube
- **Lateral folding** impinges the umbilical vesicle forming the **gut tube**

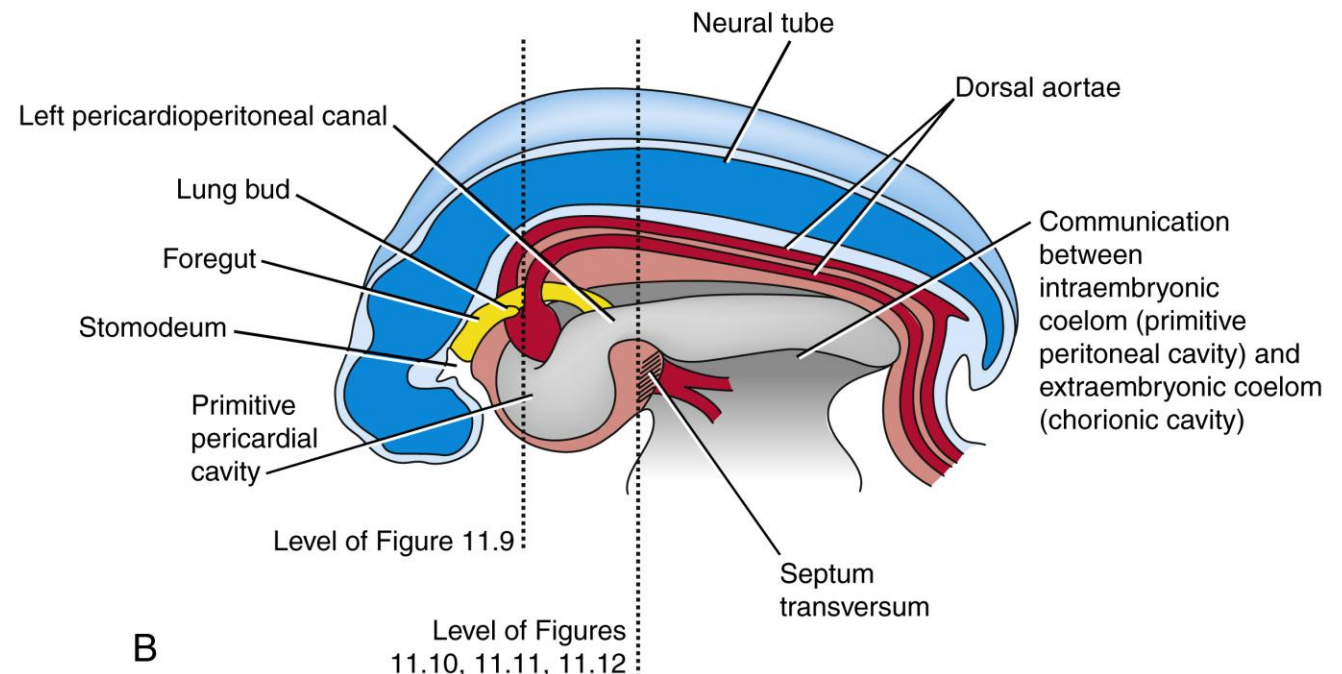
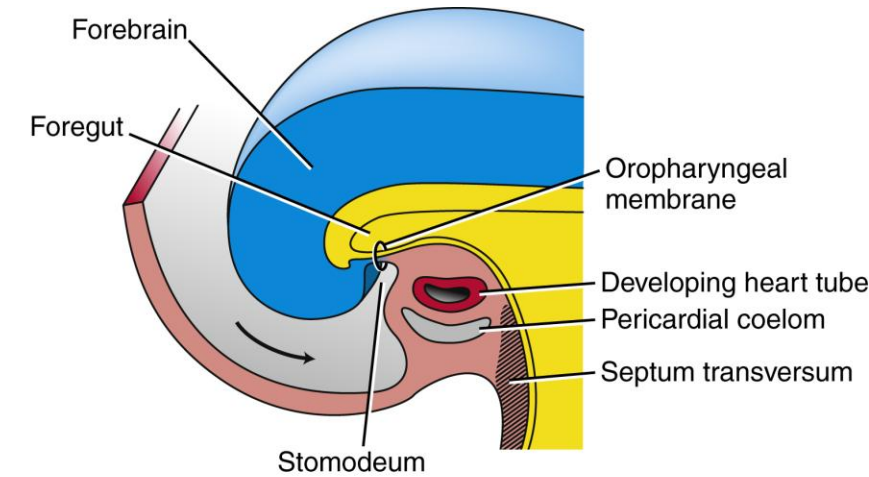


Week 4 – The embryonic body cavities and mesentery

- **Day 22** - A bar of mesoderm develops just caudal to the cranial margin of the embryonic disc = **septum transversum**
- During formation of the **head fold**, the **heart and pericardial cavity** and **septum transversum** are relocated anterior to the foregut
- Septum transversum lies just ventral to the pericardioperitoneal canals



A

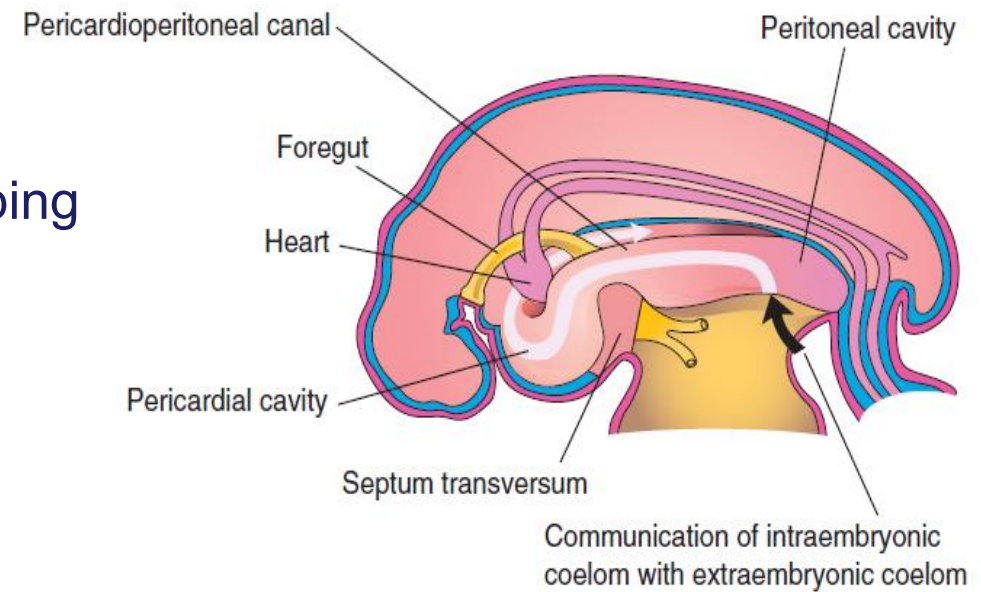


B

Level of Figures
11.10, 11.11, 11.12

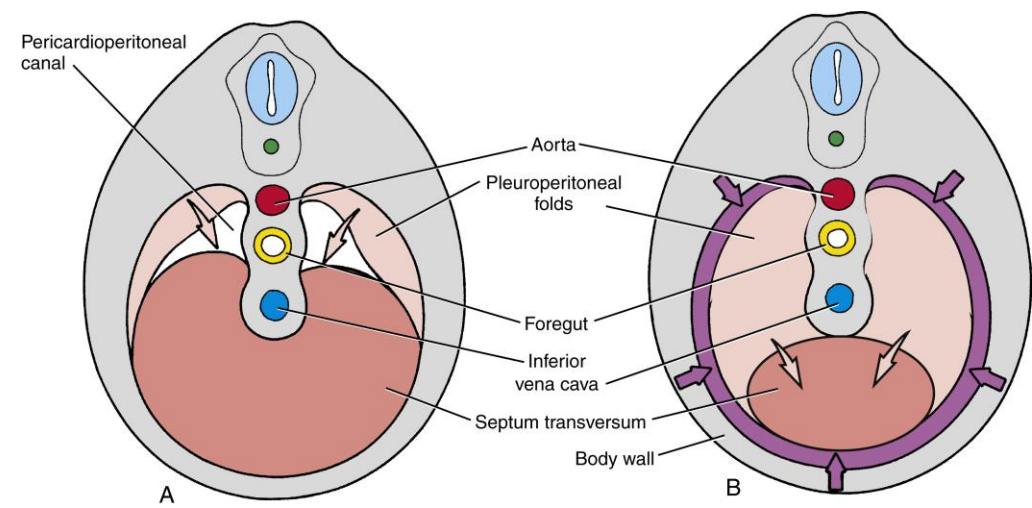
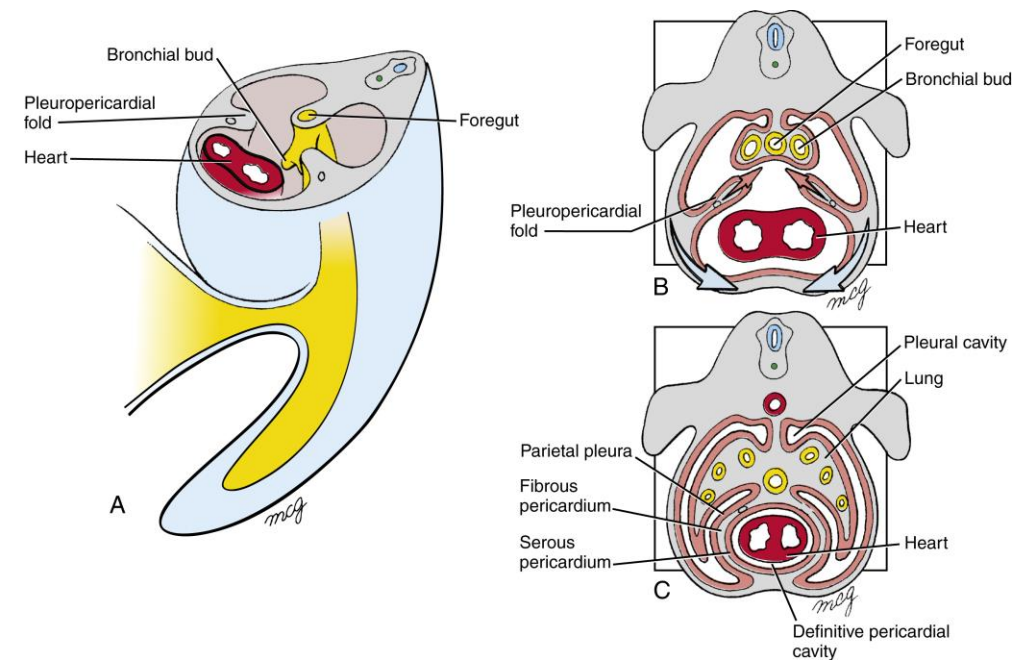
Week 4 – The embryonic body cavities and mesentery cont.

- **Intraembryonic coelom** becomes the embryonic body cavity, which is divided into three well-defined body cavities
 1. **Pericardial cavity**
 2. **Two pericardio-peritoneal canals** (will give rise to the pleural cavity)
 3. **Large peritoneal cavity**
- Somatic mesoderm gives rise to a parietal layer of mesothelium = **parietal peritoneum/pericardium/pleura**
- Splanchnic mesoderm gives rise to a visceral layer of mesothelium
= **visceral peritoneum/pericardium/pleura**
 - Give rise to the **mesentery** connecting the developing gut tube to the body wall – **contains vasculature**
 - **Dorsal** connecting the dorsal wall
 - **Ventral** connecting to the ventral wall



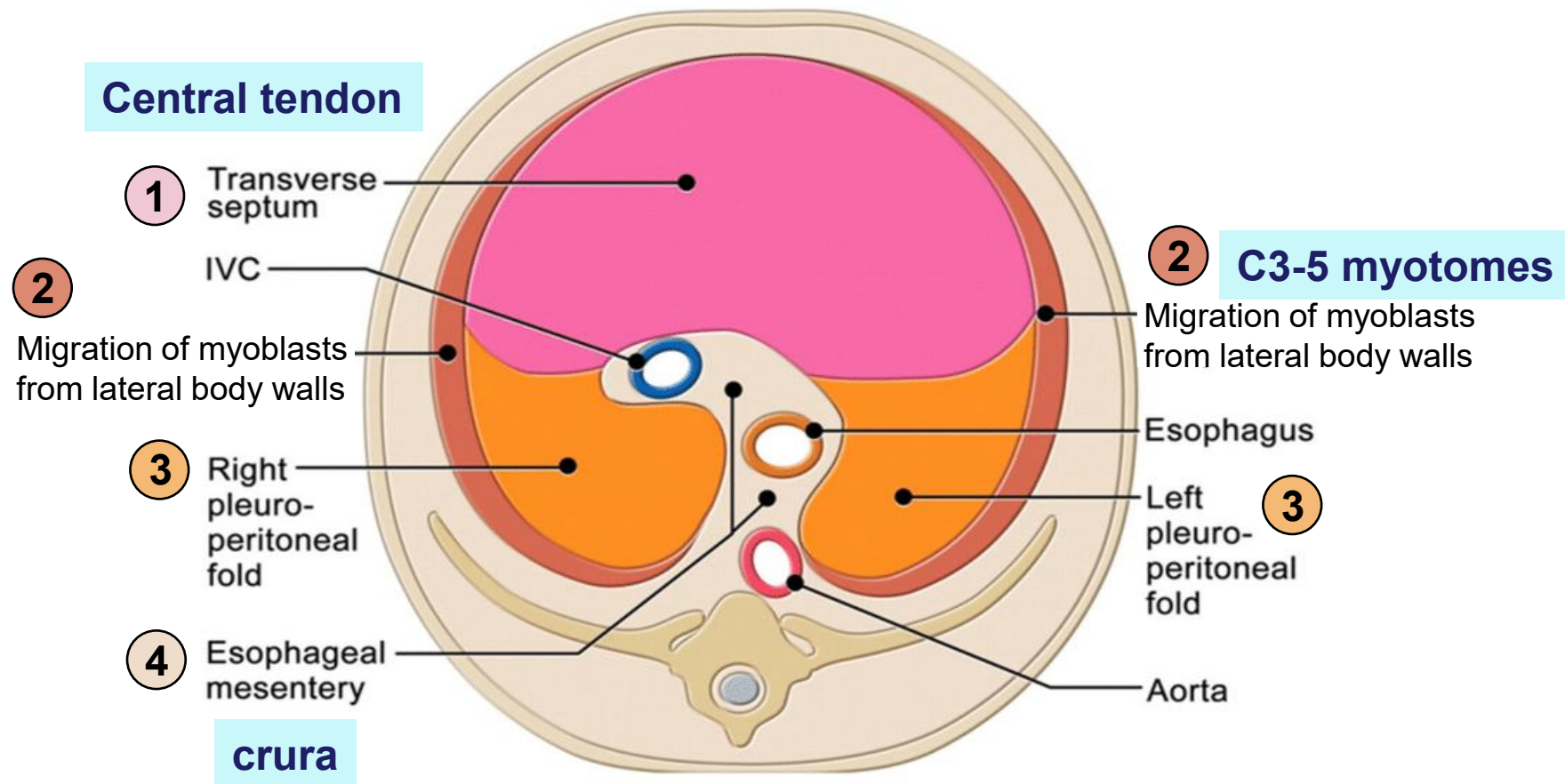
Week 5 to 7 - Partitioning of the body cavity

- Broad folds of mesenchyme appear along the lateral body walls in a coronal plane = **pleuropericardial folds**
 - Adjacent to the somites of C3 – C5 vertebrae
 - They retain the epithelial layers as they migrate resulting in a **three layered structure**
- Grow medially toward each other between the heart and lungs
- Meet and fuse with the foregut mesenchyme dividing the primitive pericardial cavity into three
 - Definitive pericardial cavity
 - 2 pleural cavities
- Pleuropericardial folds expand ventrally over the septum transversum
- Differential growth causes their location to lie dorsal to the septum transversum



Development of the Diaphragm

A composite structure that develops from **4 embryonic components**:



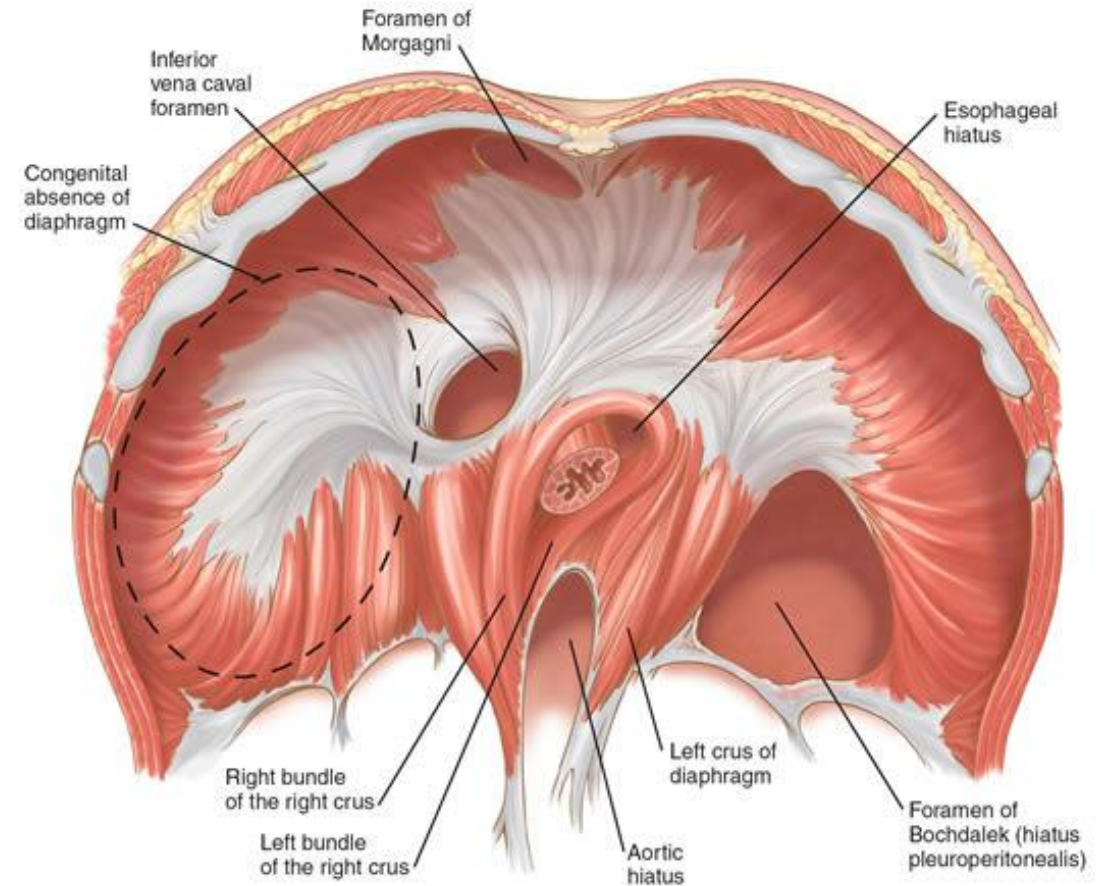
Together, they will **complete the closure** of the pleuroperitoneal openings **separating the thoracic cavity from the abdominal cavity**

Diaphragmatic Hernia

Diaphragm is formed of fusion of four structures together:

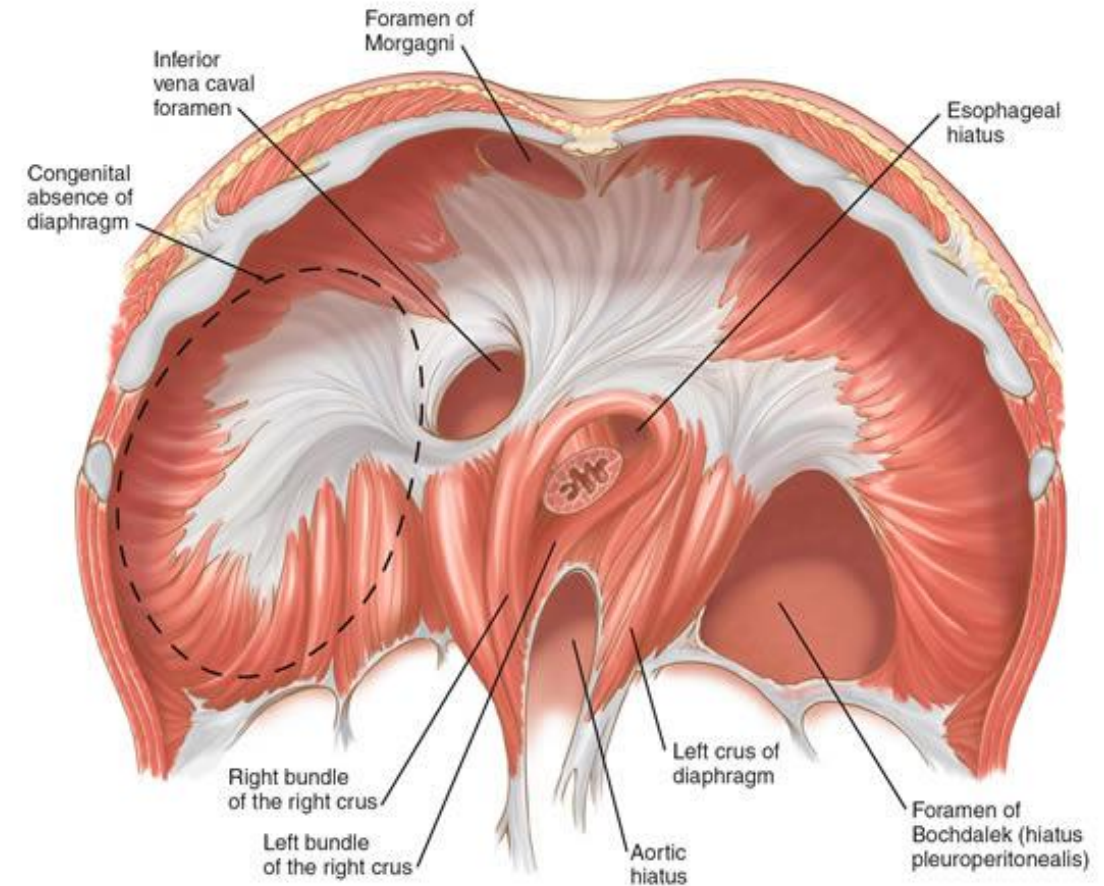
Failure of fusion of any one them will lead to hernia.

- **Morgagni's hernia:** Between the xiphoid process and the costal margin on the right
 - Also called retrosternal, parasternal
 - Rare, represent only 3% of all CDH (congenital diaphragmatic hernia)
 - Caused by a **congenital defect in the fusion of septum transversum** of the diaphragm and the costal margin
 - More common on the **right side**
 - **Allows abdominal contents to enter the thoracic cavity**



Diaphragmatic Hernia

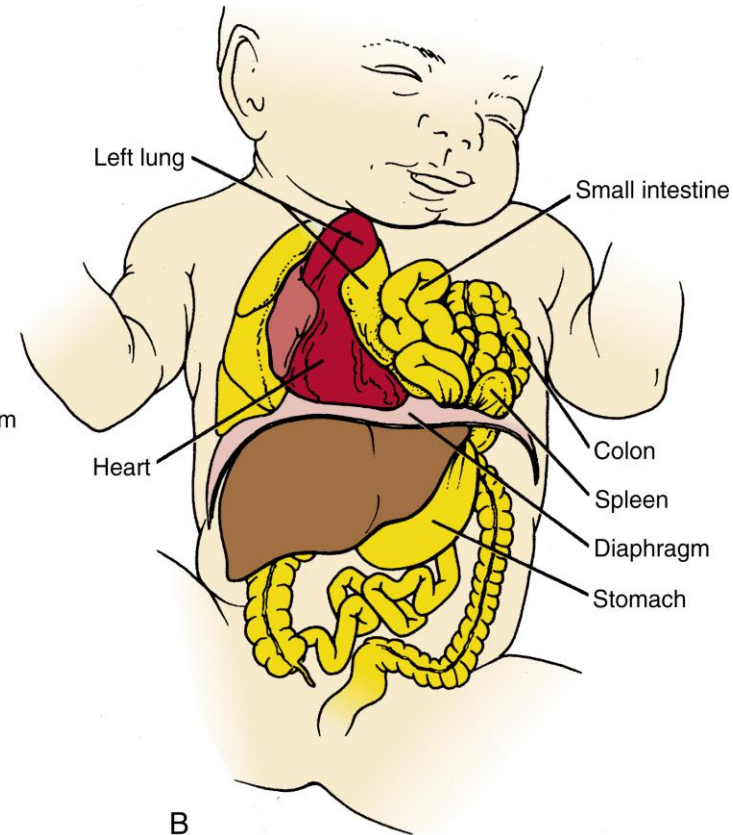
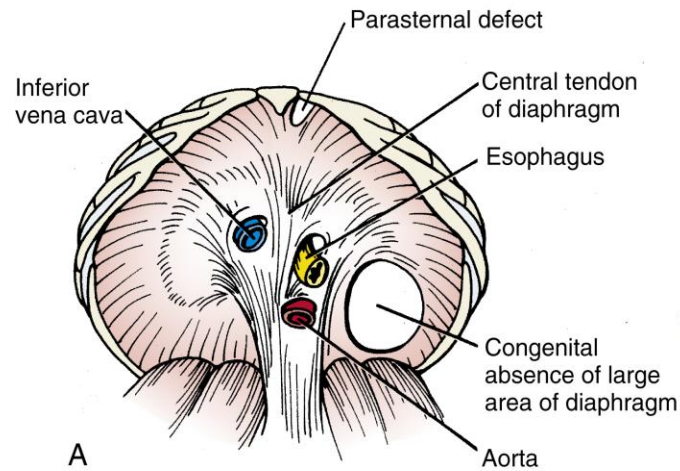
- **Bochdalek's hernia:** Posterolateral region
 - **Most common** type of hernia
 - Due to defective **formation or fusion of the pleuroperitoneal membrane.**
 - More common on the **left side**
 - **Allows abdominal contents to enter the thoracic cavity**
- **Congenital hiatal hernia:** Esophageal hiatus (opening)
 - The opening may be abnormally large allowing the fundus of the stomach to herniate into the posterior mediastinum
 - Typically cause symptoms of acid reflux



Congenital Diaphragmatic Hernia

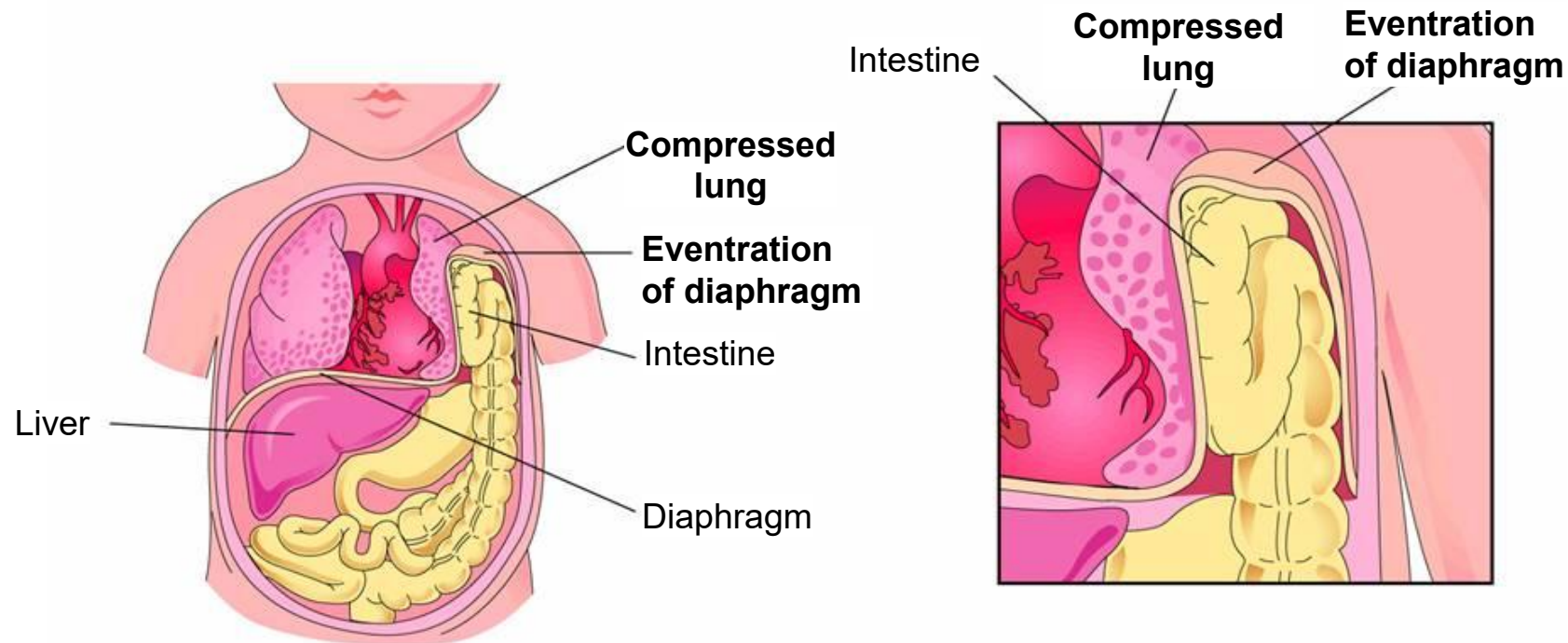
Signs and symptoms

- Abdomen is empty when lying flat as abdominal contents herniate into the thorax
- Lung is compressed & hypoplastic (**high mortality rates**) – **requires urgent surgery**



Eventration Of Diaphragm

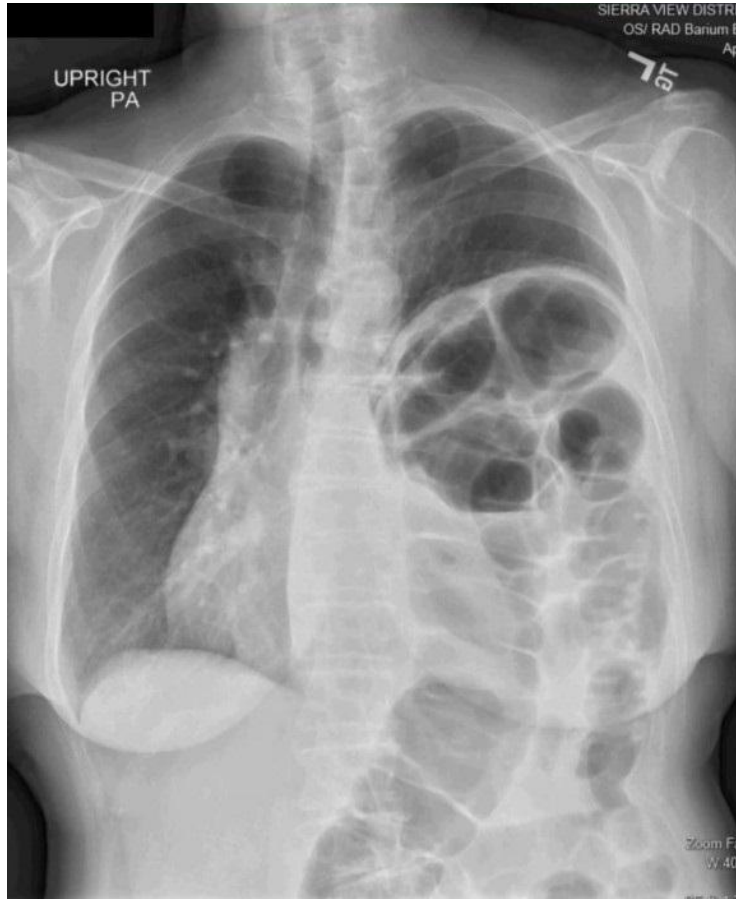
- Relatively **uncommon condition**
- Half of the diaphragm has **defective musculature**, causing it to **balloon into the thoracic cavity** as a membranous sheet, forming a **diaphragmatic pouch**
 - Consequently, the abdominal viscera are displaced superiorly into the **pocket-like outpouching of the diaphragm**
- This birth defect results mainly from **failure** of the muscular tissue from the body wall **to extend into the pleuroperitoneal membrane** on affected side
- May be asymptomatic or associated with breathing difficulty, poor feeding and poor weight gain



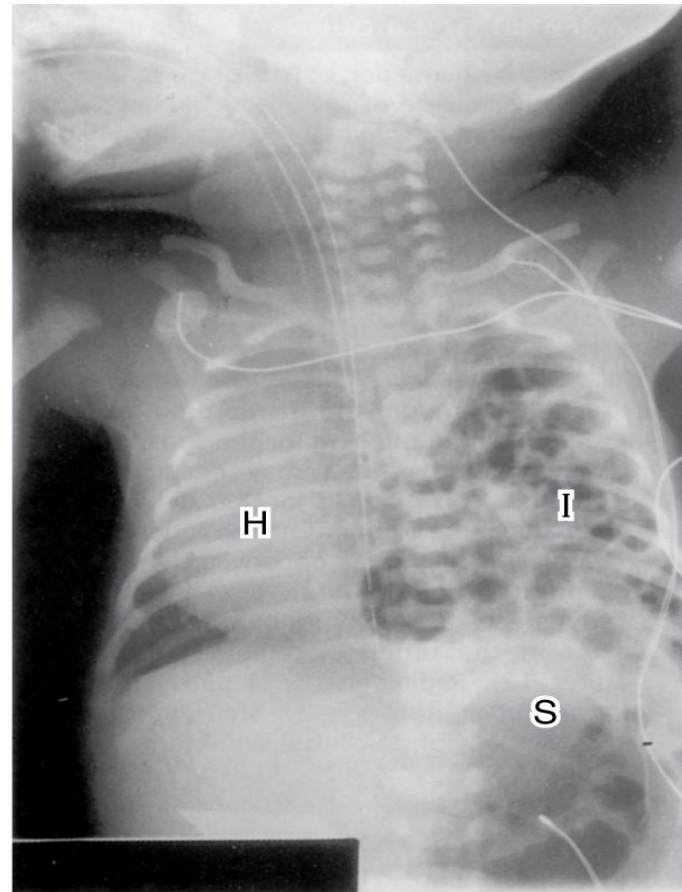
Eventration of diaphragm Vs Diaphragmatic hernia

Similar clinical presentation: Intestinal contents move into thoracic cavity

Eventration: muscular deficit and intact



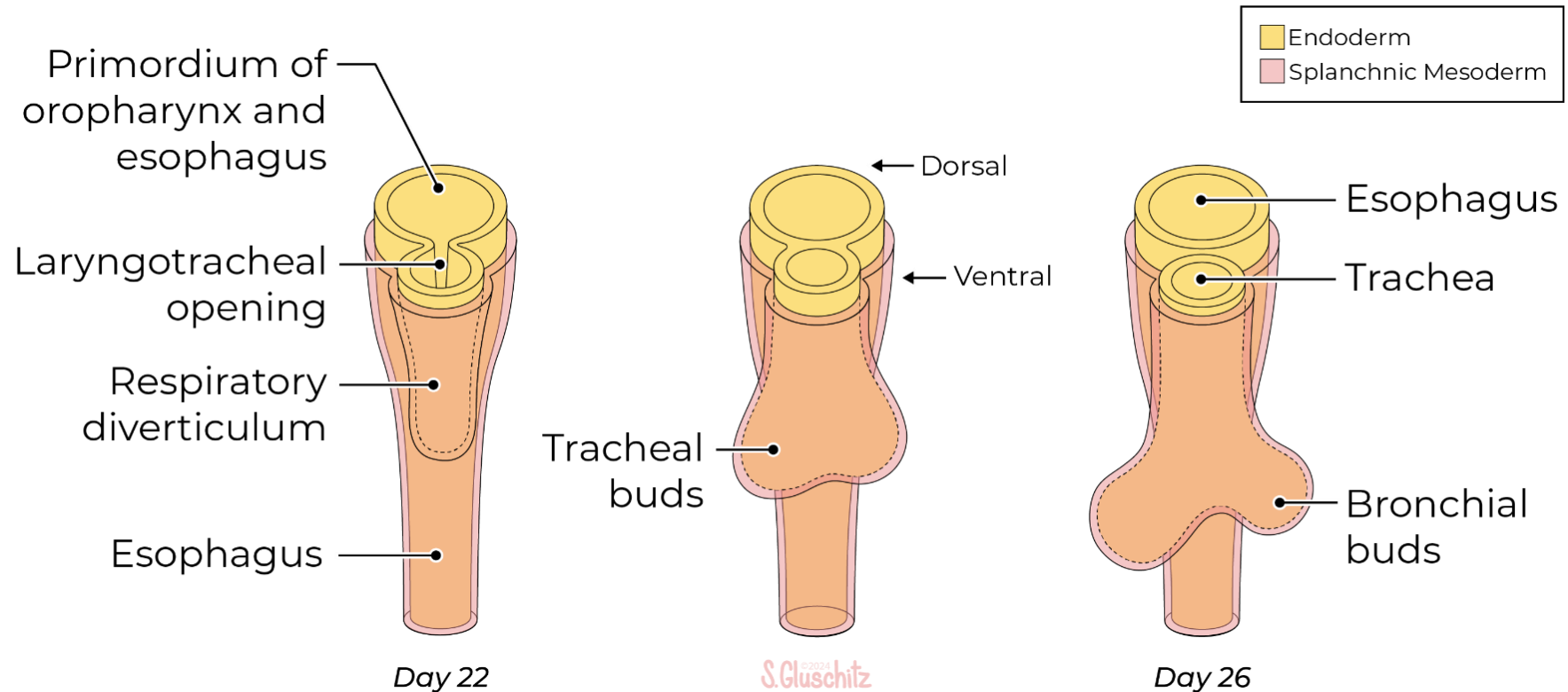
Diaphragmatic hernia:
Pleuroperitoneal membrane deficit



Week 4 – development of the respiratory diverticulum

EMBRYONIC STAGE

- Day 22 - Foregut produces a ventral outpouching = respiratory diverticulum
 - Through its development it remains ensheathed in splanchnic mesoderm
- Day 26 – 28 – the bud bifurcates into left and right primary bronchial buds
 - Gives rise to lungs and bronchi
- The stem (which remains attached to the foregut) forms the larynx and trachea

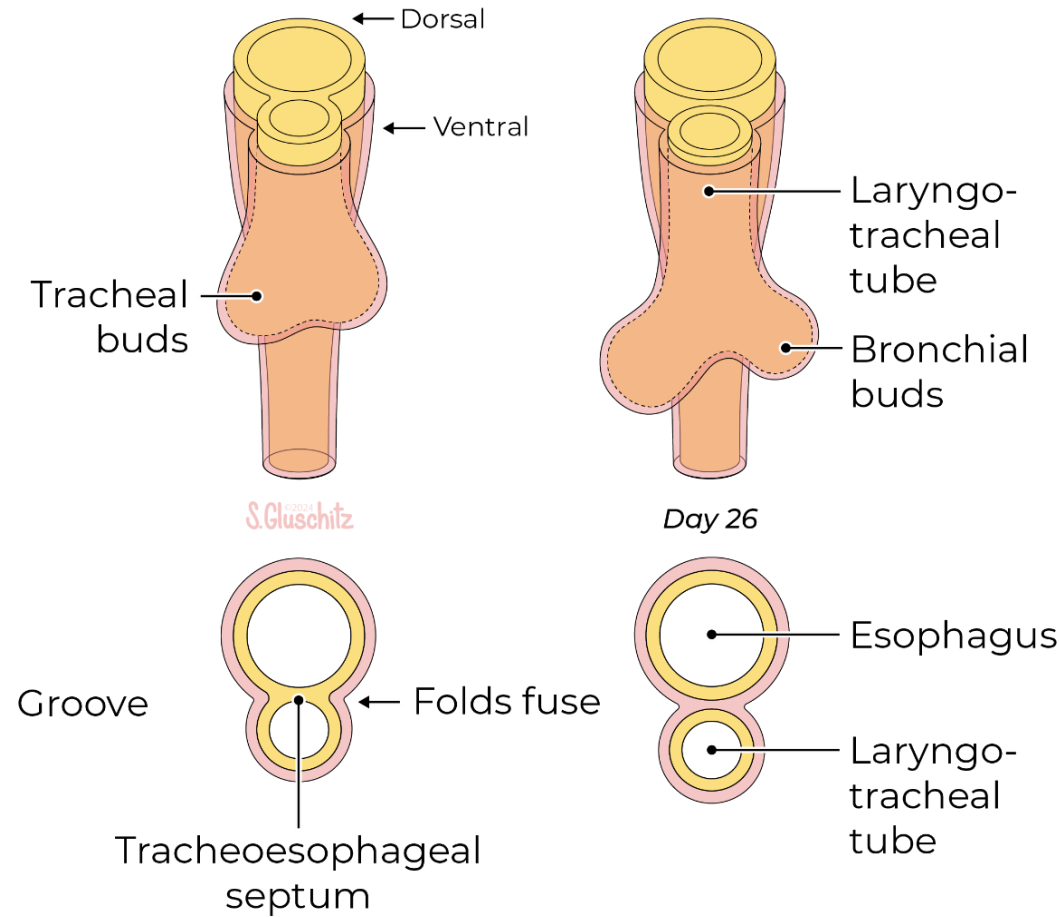
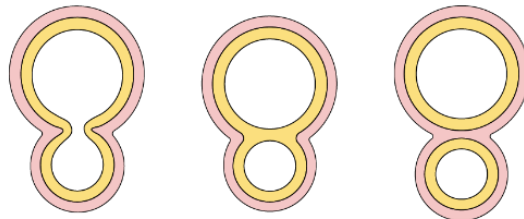


Separation of the respiratory diverticulum from the foregut

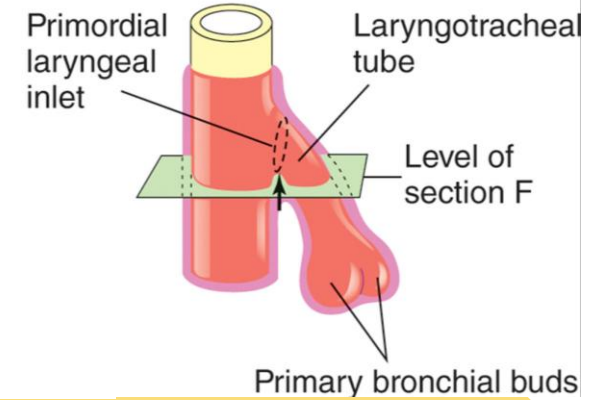
EMBRYONIC STAGE

- Week 4 - Bilateral longitudinal **tracheoesophageal folds** develop in the diverticulum
 - Grow toward each other each other
- Week 5 - fuse to form a partition = **tracheoesophageal septum**
- This septum divides the cranial portion of the foregut into:
 - **Ventral part:** laryngotracheal tube (the primordium of the larynx, trachea, bronchi, and lungs)
 - **Dorsal part:** the primordium of the oropharynx and esophagus

Cross sections day 22-26



Development of the Larynx



Epithelial lining:

Develops from the **endoderm** of the cranial end of the **laryngotracheal tube**.

Laryngeal muscles:

Develop from myoblasts in the **4th and 6th pairs** of pharyngeal arches

Laryngeal cartilages:

Develop from the cartilage's component of **4th and 6th pairs** of pharyngeal arches (from mesenchyme derived from neural crest cells)

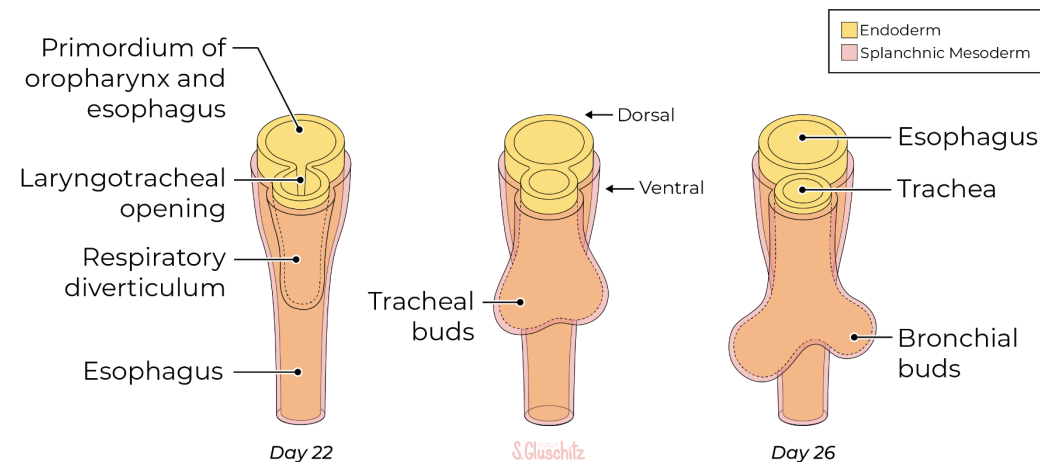
Development of the Trachea

Epithelium and glands:

Develops from the **endoderm** of the laryngotracheal tube distal to the larynx

Cartilage connective tissues and muscles:

Develop from the **splanchnic mesenchyme** surrounding the laryngotracheal tube



Embryologic Defects of the Larynx, Trachea and Esophagus

Atresia

- Complete obstruction/blockage of a passage. Here it is due to failure of recanalization of the lumen or incomplete division of the foregut

Stenosis

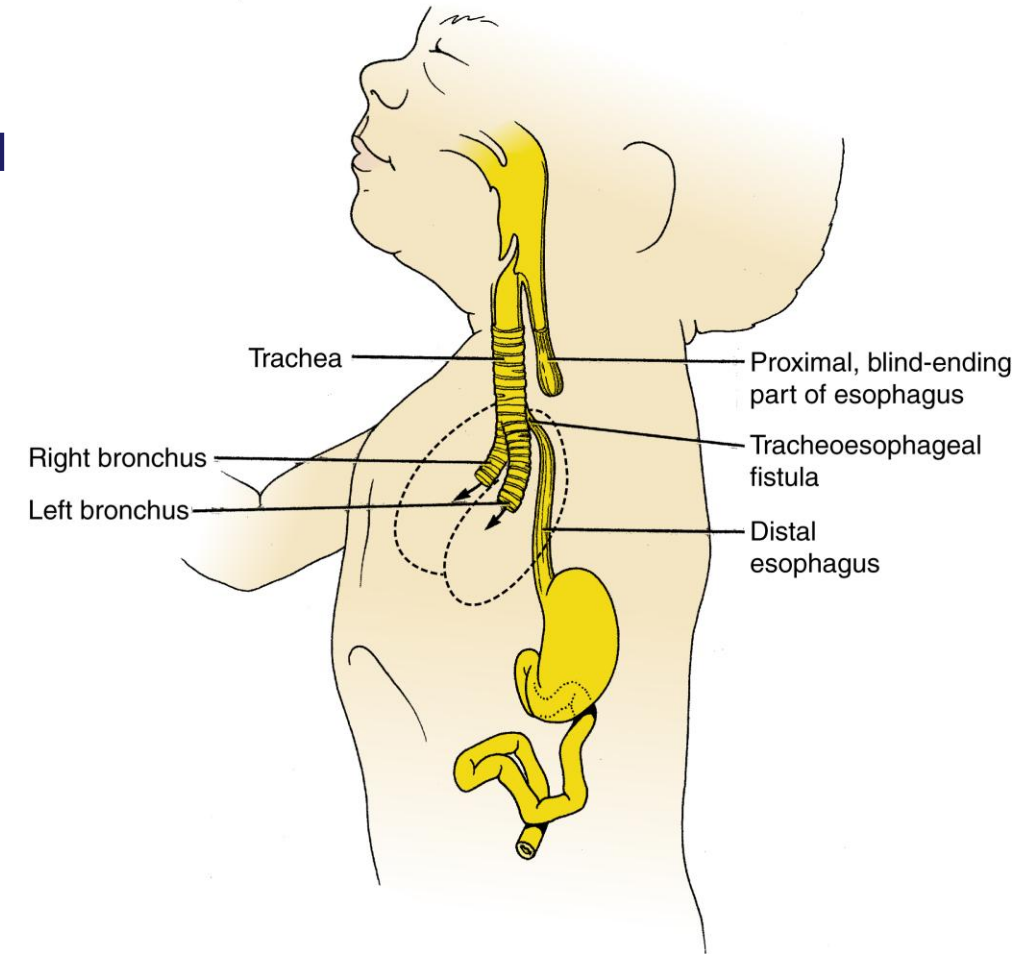
- Narrowing/partial closure of a passage. Here it is mainly due to unequal partitioning of the foregut.

Fistula

- Abnormal communication between two epithelial surfaces, due to incomplete division of a tube (e.g., incomplete division of cranial part of foregut into respiratory system and esophagus)

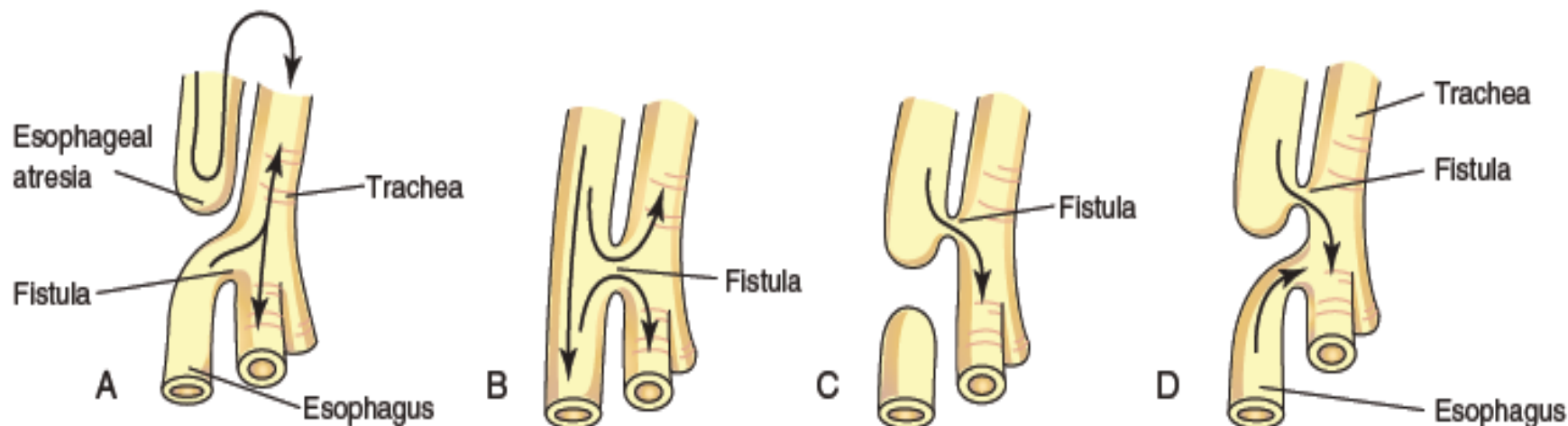
Tracheo-Esophageal Fistula (TEF)

- A TEF results from **incomplete division of the cranial part of the foregut** into respiratory and esophageal parts during the fourth week
 - result in a defective tracheoesophageal septum and a **communication between the trachea and esophagus**
- **TEF is the most common birth defect of the lower respiratory tract**
- **Polyhydramnios** is often associated with esophageal atresia
 - The excess amniotic fluid develops because fluid cannot enter the stomach and intestines for absorption and subsequent transfer through the placenta to the mother's blood for disposal

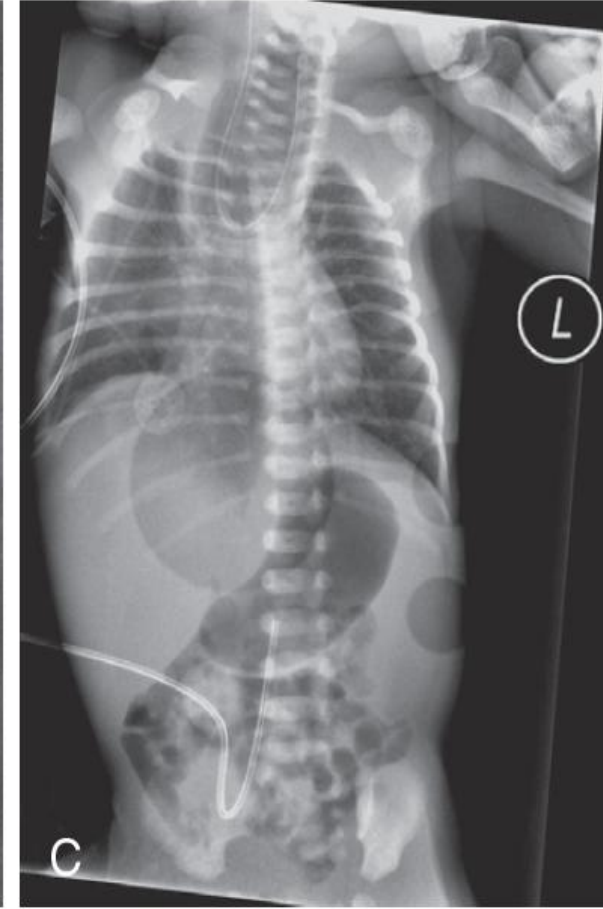
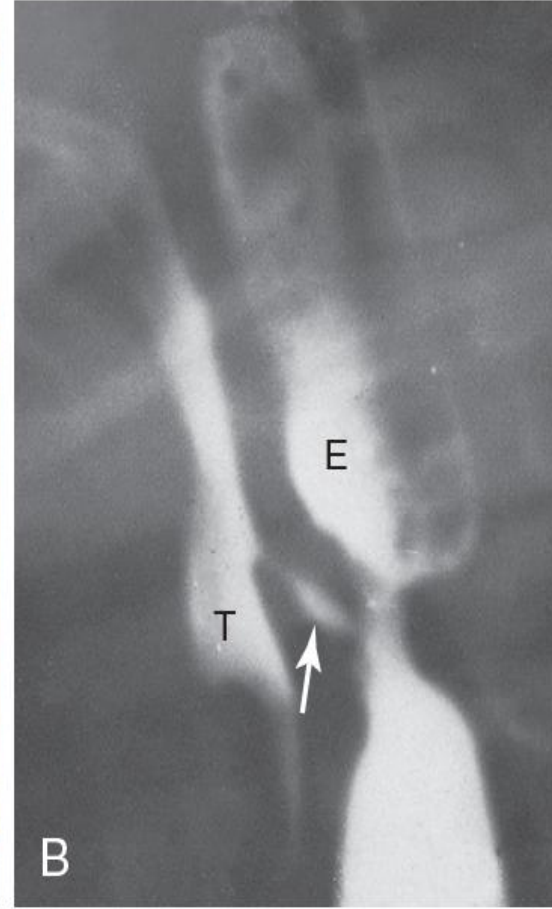
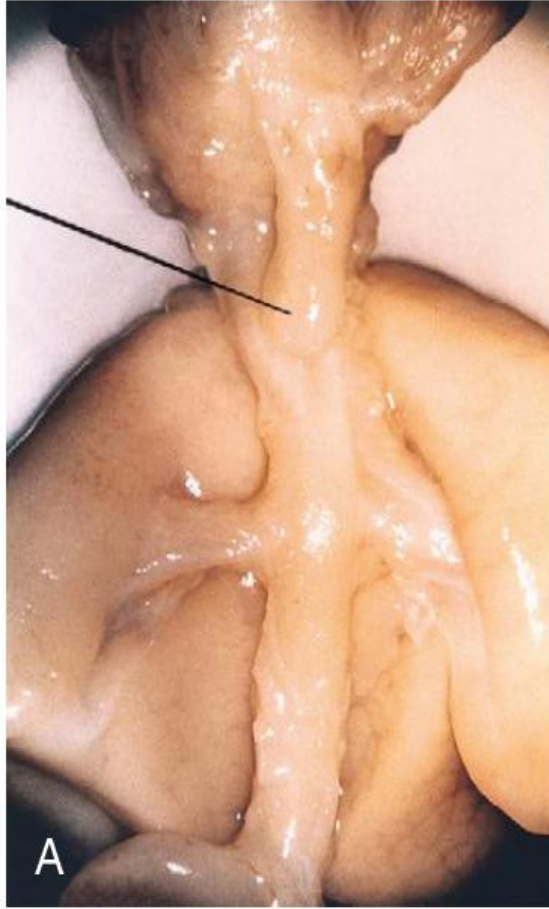


Tracheo-Esophageal Fistula (TEF)

- Four main varieties of TEF may develop
- The most common defect is for the **superior part** of the esophagus to end blindly (esophageal atresia) and for the inferior part to join the trachea near its bifurcation (image A)
 - Infants with this type of TEF and esophageal atresia cannot swallow, so they frequently drool saliva and immediately regurgitate milk when fed
- Gastric and intestinal contents may also **reflux from the stomach** through the fistula into the trachea and lungs. This refluxed acid, and in some cases bile, can cause **pneumonitis** (inflammation of the lungs)



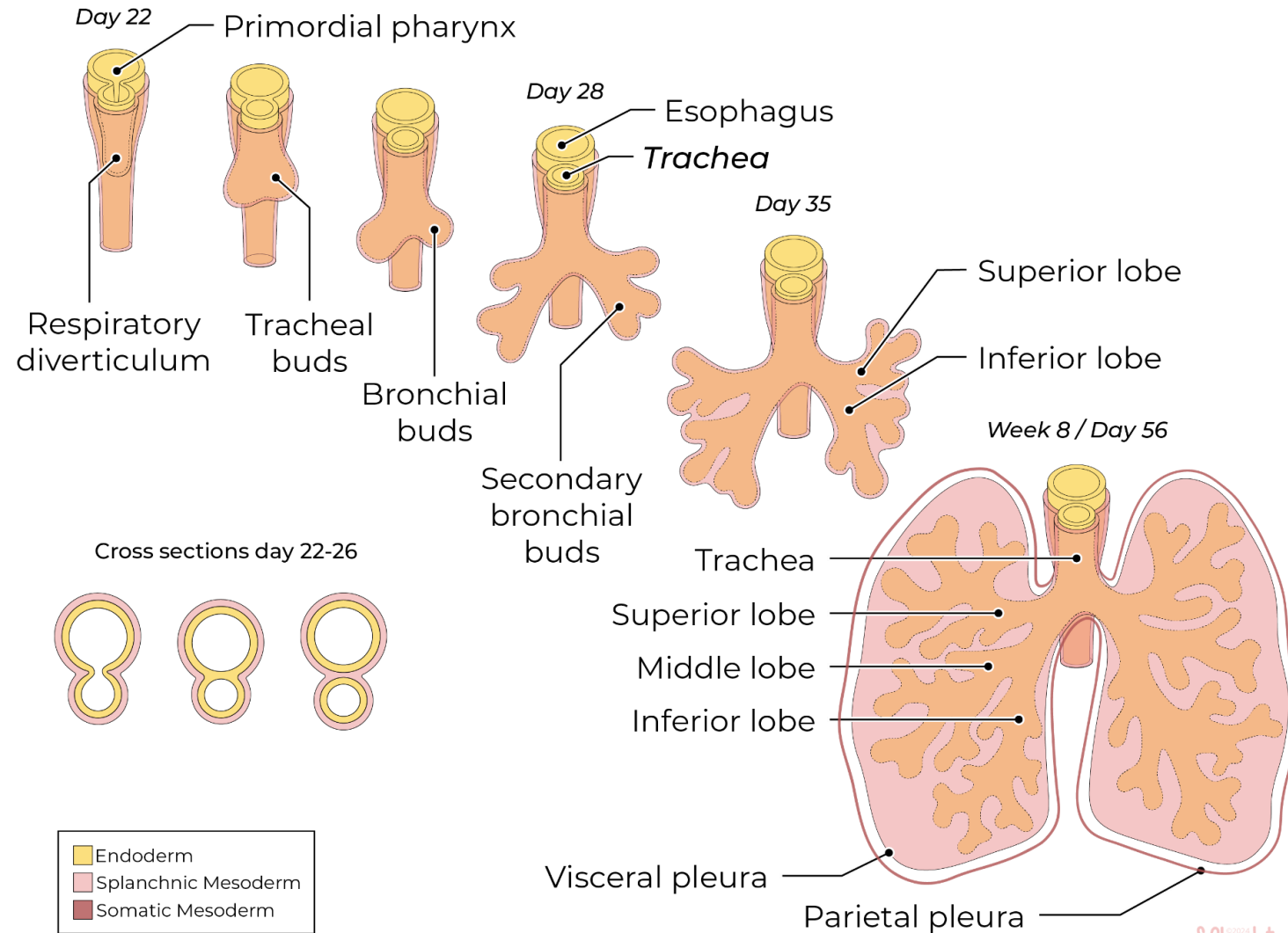
TRACHEO-ESOPHAGEAL FISTULA (TEF)



Week 5 - 16 – Continued development of the lung

PSEUDOGLANDULAR STAGE

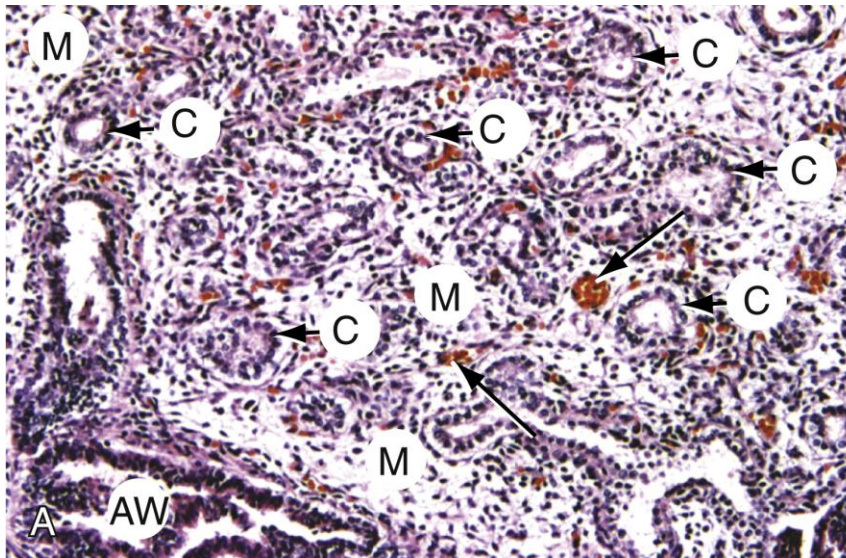
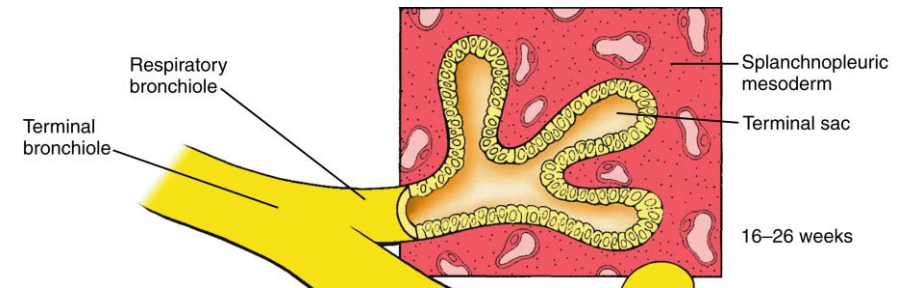
- Second branching produces
 - **3 secondary** bronchial buds on the **right**
 - **2 secondary** bronchial buds on the **left**
- Forms the future lobes
- Each secondary bronchial bud develops into 10 tertiary bronchial buds = **bronchopulmonary segments**
- They continue to grow and bud out gradually filling the pleural spaces
- **Week 16** – formation of the terminal bronchioles (16th branching)
- Fetus begin making breathing movements = causes **fluid movement** through the tubes



Week 16 – 26: Respiratory system development cont.

- Terminal bronchioles divide into two or more respiratory bronchioles
- Surrounding mesodermal tissue become highly vascularized
- Terminal sacs = **primitive alveoli**, begin forming in a craniocaudal function
- Wall tube cell start to thin
- Cells lining the respiratory structures begin to differentiate to form **type I** and **type II** pneumocytes
 - small amount of surfactant is produced
 - gas exchange is possible but very limited

CANNALICULAR STAGE

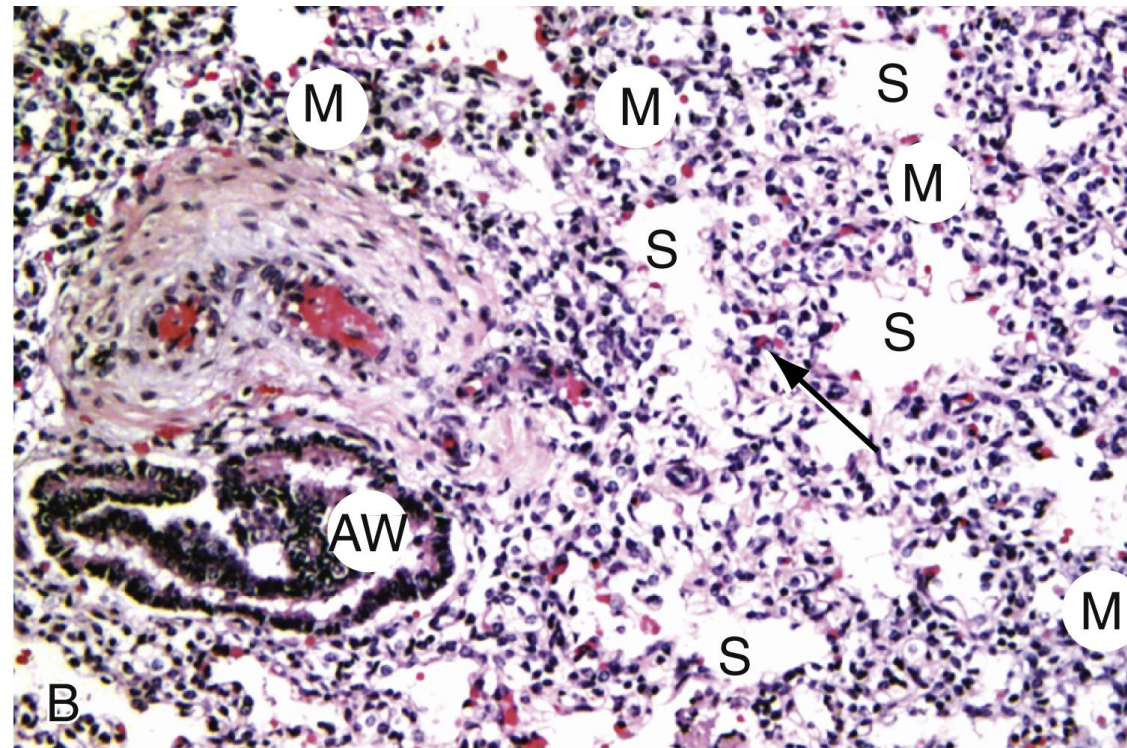


AW, airway
C, canaliculus
M, mesenchyme
Arrows, capillaries

24–38 weeks: Respiratory system development cont.

SACCULAR STAGE

- Further thinning of the tubes
- Forms numerous sacculi lined with type I and II pneumocytes
- **Surfactant is being produced by type II cells**
- Capillaries become more numerous and closely associated with the terminal sacs

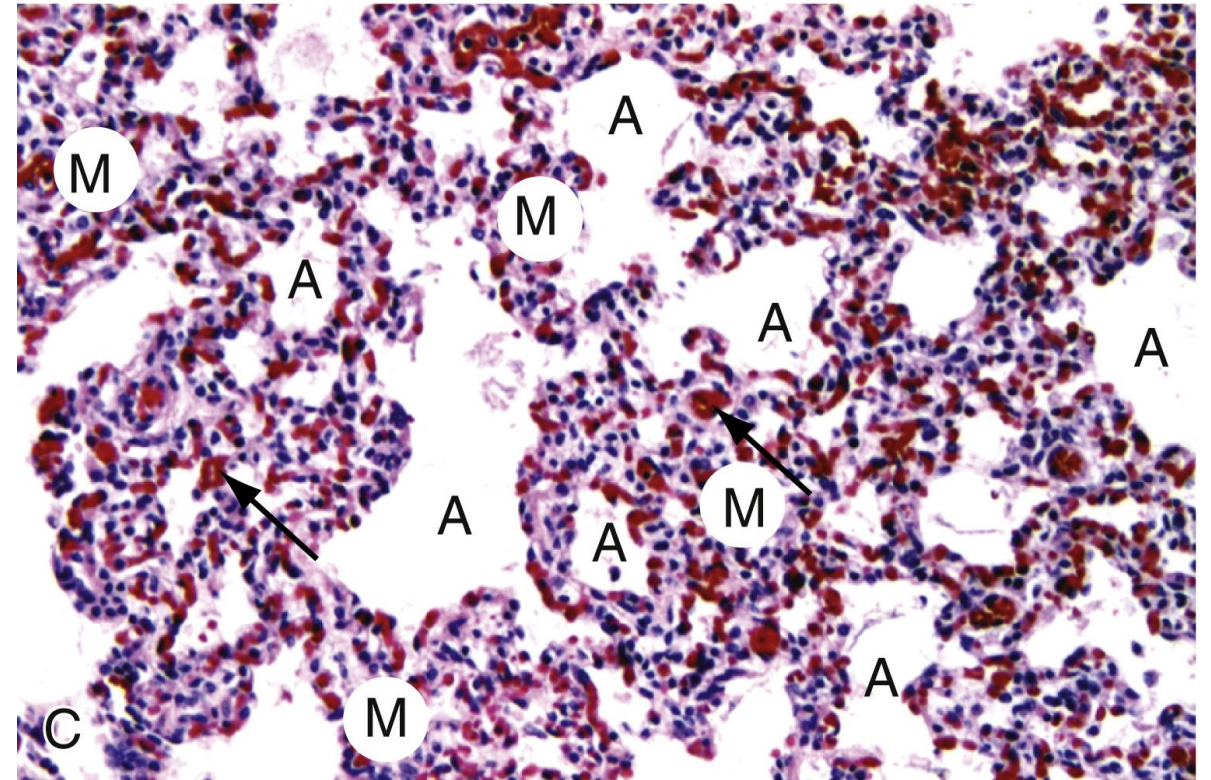
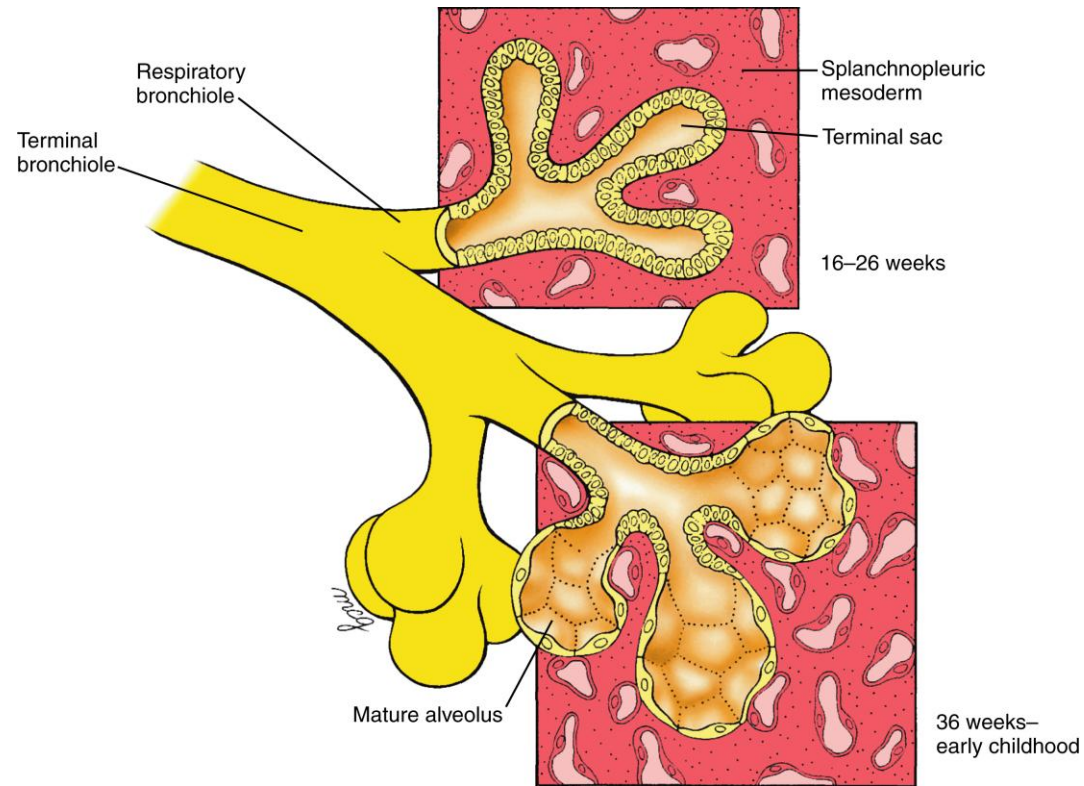


AW, airway
M, mesenchyme
S, sacculi
Arrows, capillaries

Week 36 till birth: Development of the respiratory system cont.

ALVEOLAR STAGE

- Formation of mature alveoli



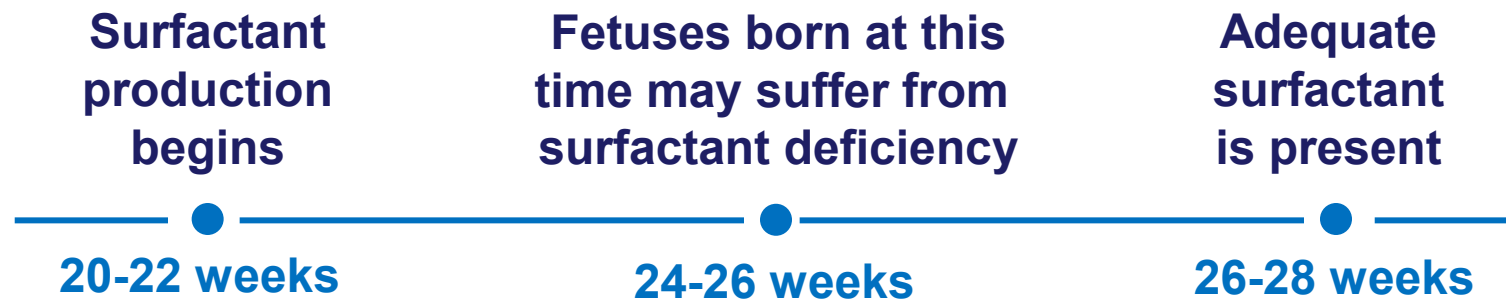
A, Alveolus
C, canaliculus
M, mesenchyme
Arrows, capillaries

Stages of Lung Development - summary

Embryonic	22 days to 6 weeks	Respiratory diverticulum arises, undergoes three initial rounds of branching, producing the primordia successively of the two lungs, the lung lobes, and the bronchopulmonary segments; the stem of the diverticulum forms the trachea and larynx
Pseudoglandular	5–16 weeks	14 more generations of branching = formation of terminal bronchioles
Canalicular	16–26 weeks	Each terminal bronchiole divides into two or more respiratory bronchioles. Respiratory vasculature begins to develop. The lung epithelium begins to differentiate into specialized cell types (ciliated, bronchiolar secretory, and neuroendocrine cells proximally and precursors of the alveolar type II and I cells distally)
Saccular	24–38 weeks	Respiratory bronchioles subdivide to produce terminal sacs (primitive alveoli). Terminal sacs continue to be produced until well into childhood
Alveolar	36 weeks to early adulthood	Alveoli mature. Continued septation of alveoli as the lungs grow into early adulthood.

Surfactant

- Surface-active agent
- Lipoprotein produced by **Type II alveolar cells**
- **Reduces the surface tension**, so facilitates expansion of the terminal sacs and prevent collapse
- Production **begins at 20 to 22 weeks**
- Fetuses born at **24 to 26 weeks** after fertilization may survive if given intensive care; however, they may suffer from **respiratory distress** because of **surfactant deficiency**
- By **weeks 26-28** – adequate terminal sacs & surfactant are present to permit survival of a prematurely born infant



Neonatal Respiratory Distress Syndrome

- **Approx. 2% of live births**; and those born prematurely are the most susceptible
- RDS is also known as **hyaline membrane disease**
- An estimated 30% of all neonatal diseases result from RDS or its complications
- Symptoms: **rapid, labored breathing shortly after birth**
- Causes:
 - **Surfactant deficiency**
 - **Prolonged intrauterine asphyxia** (impaired or absent exchange of oxygen and carbon dioxide) may produce irreversible changes in the type II alveolar cells, making them incapable of producing surfactant
- **Treatment** includes **supplementary oxygen** and artificial surfactant —more than 90% of neonates with RDS survive
- **Corticosteroids** are potent **stimulators** of fetal surfactant production and may be given to the mother if early delivery is a risk

Respiratory development and changes after birth

Changes in Respiratory system at birth

- First breath **expands lungs**, pulmonary **resistance decreases**, pulmonary **blood flow increases**
- Airspaces of the fetal lung are filled with fetal lung fluid so need to be cleared;
 - By pressure on the thorax during vaginal delivery (**mechanical expulsion**)
 - Into the **pulmonary capillaries, pulmonary arteries, veins** and lymphatic vessels
- Fetal breathing become autonomous and continuous (Transformation of the lungs into gas-exchanging organ)
- Establishment of parallel pulmonary and systemic circulations

Continued development

- Additional **terminal sacs develop until about 2 years** of age
- Septation subdivides the alveoli and continues **into early adulthood**
 - Septum adds smooth muscle and capillaries